

Enterprise Server Deployment and Operating System Installation
Week 3 Portfolio Project ITEP 414 – System Administration and Maintenance

1. Project Overview

This project records the setup of a clean Ubuntu Server virtual machine for ABC Startup Solutions. The server is designed to facilitate file sharing, remote management, database hosting, web hosting, and internal services. The guide is crafted to enable another junior system administrator to replicate the deployment following the same standards.

2. Objectives

- Explain the purpose of an operating system in enterprise environments.
- Differentiate BIOS and UEFI firmware.
- Explain the computer boot process.
- Install Ubuntu Server in a virtual machine.
- Configure server settings and SSH.
- Verify server functionality.
- Document the deployment professionally.

3. Software Requirements

- Oracle VirtualBox or VMware Workstation
- Ubuntu Server LTS ISO
- Windows Server Evaluation ISO for the bring-home activity

4. Virtual Machine Specifications

Component	Required Specification
VM Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 virtual processors
Storage	40 GB VDI/VMDK
Network	NAT (or Bridged if instructed)
Optical Drive	Ubuntu Server ISO

5. Step-by-Step Ubuntu Server Installation

1. Create a new virtual machine named Ubuntu-Server-Week03.
2. Assign 4 GB RAM, 2 virtual CPUs, and a 40 GB virtual disk.
3. Attach the Ubuntu Server LTS ISO to the optical drive.
4. Set networking to NAT unless instructed otherwise.
5. Start the VM and select English as the language.
6. Select the appropriate keyboard layout, such as English (US).
7. Accept DHCP and record the assigned IP address.
8. Set the hostname to server01.
9. Create a non-root administrative user using your own name/username and a strong password.
10. Select Guided – Use Entire Disk for storage.

11. Record the partition scheme, file system, and disk size shown by the installer.
12. Enable Install OpenSSH Server. 13. Do not install additional packages unless instructed.
14. Complete the installation, reboot, and remove the ISO if prompted.

6. Configuration Details

Hostname: server01

Network: DHCP

Administrative access: non-root user with sudo privileges

Remote administration: OpenSSH Server

Storage: Guided – Use Entire Disk

7. Server Verification Procedures

After logging in, run the following commands and capture screenshots.

`hostname`

Expected result: server01

`ip addr`

Expected result: the assigned network interface and IP address.

`ping -c 4 google.com`

Expected result: successful replies with packet statistics.

`sudo apt update` `sudo apt upgrade -y`

Expected result: package lists and installed packages are updated without critical errors.

`systemctl status ssh`

Expected result: SSH service displays active (running).